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Jonah drove the first half of a 100-mile trip in x hours and the second half in y hours. Which of the following is equal to Jonah's average speed, in miles per hour, for the entire trip?

A. $\frac{50}{x+y}$

B. $\frac{100}{x+y}$

C. $\frac{25}{x}+\frac{25}{y}$

D. $\frac{50}{x}+\frac{50}{y}$

E. $\frac{100}{x}+\frac{100}{y}$

•

If the amount of federal estate tax due on an estate valued at \$1.35 million is \$437,000 plus 43 percent of the value of the estate in excess of \$1.25 million, then the federal tax due is approximately what percent of the value of the estate?

A. 30%

B. 35%

C. 40%

D. 45%

E. 50%

$$7x+6y \leq 38{,}000$$
$$4x+5y \leq 28{,}000$$

•

A manufacturer wants to produce x balls and y boxes. Resource constraints require that x and y satisfy the inequalities shown. What is the maximum number of balls and boxes combined that can be produced given the resource constraints?

A. 5,000

B. 6,000

C. 7,000

D. 8,000

E. 10,000

•

If $\frac{3}{10^4} = x\%$, then $x =$

A. 0.3

B. 0.03

C. 0.003

D. 0.0003

E. 0.00003

•

What is the remainder when 3^{24} is divided by 5?

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What is the remainder when 3 is divided by 3?

A. 0

B. 1

C. 2

D. 3

E. 4

•

José has a collection of 100 coins, consisting of nickels, dimes, quarters, and half-dollars. If he has a total of 35 nickels and dimes, a total of 45 dimes and quarters, and a total of 50 nickels and quarters, how many half-dollars does he have?

A. 15

B. 20

C. 25

D. 30

E. 35

•

David used part of $100,000 to purchase a house. Of the remaining portion, he invested $\frac{1}{3}$ of it at 4 percent simple annual interest and $\frac{2}{3}$ of it at 6 percent simple annual interest. If after a year the income from the two investments totaled $320, what was the purchase price of the house?

A. $96,000

B. $94,000

C. $88,000

D. $75,000

E. $40,000

•

A certain manufacturer sells its product to stores in 113 different regions worldwide, with an average (arithmetic mean) of 181 stores per region. If last year these stores sold an average of 51,752 units of the manufacturer's product per store, which of the following is closest to the total number of units of the manufacturer's product sold worldwide last year?

A. 10^6

B. 10^7

C. 10^8

D. 10^9

E. 10^{10}

•

Andrew started saving at the beginning of the year and had saved $240 by the end of the year. He continued to save and by the end of 2 years had saved a total of $540. Which of the following is closest to the percent increase in the amount Andrew saved during the second year compared to the amount he saved during the first year?

A. 11%

B. 25%

C. 44%

D. 56%

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D. 56%

E. 125%

•

If x is a positive integer, r is the remainder when x is divided by 4, and R is the remainder when x is divided by 9, what is the greatest possible value of $r^2 + R$?

A. 25

B. 21

C. 17

D. 13

E. 11

•

Each of the nine digits 0, 1, 1, 4, 5, 6, 8, 8, and 9 is used once to form 3 three-digit integers. What is the greatest possible sum of the 3 integers?

A. 1,752

B. 2,616

C. 2,652

D. 2,775

E. 2,958

•

Given that $1^2 + 2^2 + 3^2 + \ldots + 10^2 = 385$, what is the value of $3^2 + 6^2 + 9^2 + \ldots + 30^2$?

A. 1,155

B. 1,540

C. 1,925

D. 2,310

E. 3,465

•

If water is leaking from a certain tank at a constant rate of 1,200 milliliters per hour, how many <u>seconds</u> does it take for 1 milliliter of water to leak from the tank?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. 2

D. 3

E. 20

•

When the positive integer k is divided by the positive integer n, the remainder is 11. If $\frac{k}{n} = 81.2$, what is the value of n?

A. 9

B. 20

C. 55

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D. 70

E. 81

•

The total area of a certain continent is approximately 3.8×10^{16} square inches. Which of the following is closest to the area of the continent, in square miles? (1 square mile is approximately 4.0×10^{9} square inches.)

A. 6.7×10^5

B. 2.0×10^6

C. 9.5×10^6

D. 1.1×10^7

E. 9.5×10^8

•

If $xy = 1$, what is the value of $\frac{2(x+y)^2}{2(x-y)^2}$?

A. 2

B. 4

C. 8

D. 16

E. 32

•

Of the 20 members of a kitchen crew, 17 can use the meat-cutting machine, 18 can use the bread-slicing machine, and 15 can use both machines. If one member of the crew is to be chosen at random, what is the probability that the member chosen will be someone who cannot use either machine?

A. 0

B. $\frac{1}{10}$

C. $\frac{1}{7}$

D. $\frac{1}{4}$

E. $\frac{1}{3}$

•

Which of the following is an integer?

I. $\frac{12!}{6!}$

II. $\frac{12!}{8!}$

III. $\frac{12!}{7!5!}$

A. I only

B. II only

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C. III only

D. I and II only

E. I, II, and III

•

The function f is defined for all nonzero x by the equation $f(x) = x - \frac{1}{x}$. If $x \neq 0$, which of the following equals $f\left(\frac{1}{x}\right)$?

A. $f(x)$

B. $f(-x)$

C. $f\left(-\frac{1}{x}\right)$

D. $\frac{1}{f(x)}$

E. $-\frac{1}{f(x)}$

•

In the arithmetic sequence $t_1, t_2, t_3, \cdots, t_n, \cdots, t_1 = 23$ and $t_n = t_{n-1} - 3$ for each $n > 1$. What is the value of n when $t_n = -4$?

A. −1

B. 7

C. 10

D. 14

E. 20

•

How many seconds will it take for a car that is traveling at a constant rate of 45 miles per hour to travel a distance of 220 yards? (1 mile = 1,760 yards)

A. 8

B. 9

C. 10

D. 11

E. 12

•

A store's selling price of $2,240 for a certain computer would yield a profit of 40 percent of the store's cost for the computer. What selling price would yield a profit of 50 percent of the computer's cost?

A. $2,400

B. $2,464

C. $2,650

D. $2,732

E. $2,800

•

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If a certain coin is flipped, the probability that the coin will land heads up is $\frac{1}{2}$. If the coin is flipped 5 times, what is the probability that it will land heads up on the first 3 flips and not on the last 2 flips?

A. $\frac{3}{5}$

B. $\frac{1}{2}$

C. $\frac{1}{5}$

D. $\frac{1}{8}$

E. $\frac{1}{32}$

•

The operation $\otimes$ is defined for all nonzero numbers a and b by $a \otimes b = \frac{a}{b} - \frac{b}{a}$. If x and y are nonzero numbers, which of the following statements must be true?

I. $x \otimes xy = x(1 \otimes y)$

II. $x \otimes y = -(y \otimes x)$

III. $\frac{1}{x} \otimes \frac{1}{y} = y \otimes x$

A. I only

B. II only

C. III only

D. I and II

E. II and III

•

If each term in the sum $a_1 + a_2 + \cdots + a_n$ is either 7 or 77 and the sum equals 350, which of the following could be equal to n?

A. 38

B. 39

C. 40

D. 41

E. 42

•

(1.00001)(0.99999) – (1.00002)(0.99998) =

A. 0

B. 10^{-10}

C. $3(10^{-10})$

D. 10^{-5}

E. $3(10^{-5})$

•

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•

A certain rectangular photograph has an area of $6\frac{1}{2}$ square feet. What is the area of the photograph in square inches? (1 foot = 12 inches)

A. 36

B. 78

C. 432

D. 864

E. 936

•

The toll for crossing a certain bridge is \$0.75 each crossing. Drivers who frequently use the bridge may instead purchase a sticker each month for \$13.00 and then pay only \$0.30 each crossing during that month. If a particular driver will cross the bridge twice on each of x days next month and will not cross the bridge on any other day, what is the least value of x for which this driver can save money by using the sticker?

A. 14

B. 15

C. 16

D. 28

E. 29

Questions 152 to 207 — Difficulty: Hard

152. Two numbers differ by 2 and sum to S. Which of the following is the greater of the numbers in terms of S?

A. $\frac{S}{2} - 1$

B. $\frac{S}{2}$

C. $\frac{S}{2} + \frac{1}{2}$

D. $\frac{S}{2} + 1$

E. $\frac{S}{2} + 2$

153. If m is an integer and $m = 10^{32} - 32$, what is the sum of the digits of m?

A. 257

B. 264

C. 275

D. 284

E. 292

154. In a numerical table with 10 rows and 10 columns, each entry is either a 9 or a 10. If the number of 9s in the nth row is $n - 1$ for each n from 1 to 10, what is the average (arithmetic mean) of all the numbers in the table?

A. 9.45

B. 9.50

C. 9.55

D. 9.65

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E. 9.70

155. In 2004, the cost of 1 year-long print subscription to a certain newspaper was \$4 per week. In 2005, the newspaper introduced a new rate plan for 1 year-long print subscription: \$3 per week for the first 40 weeks of 2005 and \$2 per week for the remaining weeks of 2005. How much less did 1 year-long print subscription to this newspaper cost in 2005 than in 2004?

A. \$64

B. \$78

C. \$112

D. \$144

E. \$304

156. A positive integer n is a perfect number provided that the sum of all the positive factors of n, including 1 and n, is equal to $2n$. What is the sum of the reciprocals of all the positive factors of the perfect number 28?

A. $\frac{1}{4}$

B. $\frac{56}{27}$

C. 2

D. 3

E. 4

157. The infinite sequence $a_1, a_2, \ldots, a_n, \ldots$ is such that $a_1 = 2$, $a_2 = -3$, $a_3 = 5$, $a_4 = -1$, and $a_n = a_{n-4}$ for $n > 4$. What is the sum of the first 97 terms of the sequence?

A. 72

B. 74

C. 75

D. 78

E. 80

158. The sequence $a_1, a_2, \ldots a_n, \ldots$ is such that $a_n = 2a_{n-1} - x$ for all positive integers $n \geq 2$ and for a certain number x. If $a_5 = 99$ and $a_3 = 27$, what is the value of x?

A. 3

B. 9

C. 18

D. 36

E. 45

159. In a certain medical survey, 45 percent of the people surveyed had the type A antigen in their blood and 3 percent had both the type A antigen and the type B antigen. Which of the following is closest to the percent of those with the type A antigen who also had the type B antigen?

A. 1.35%

B. 6.67%

C. 13.50%

D. 15.00%

E. 42.00%

160. On a certain transatlantic crossing, 20 percent of a ship's passengers held round-trip tickets and also took their cars aboard the ship. If 60 percent of the passengers with round-trip tickets did <u>not</u> take their cars aboard the ship, what percent of the ship's passengers held round-trip tickets?

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ship, what percent of the ship's passengers held round-trip tickets?

A. $33\frac{1}{3}\%$

B. 40%

C. 50%

D. 60%

E. $66\frac{2}{3}\%$

161. If x and k are integers and $(12^x)(4^{2x+1}) = (2^k)(3^2)$, what is the value of k?

A. 5

B. 7

C. 10

D. 12

E. 14

162. If S is the sum of the reciprocals of the 10 consecutive integers from 21 to 30, then S is between which of the following two fractions?

A. $\frac{1}{3}$ and $\frac{1}{2}$

B. $\frac{1}{4}$ and $\frac{1}{3}$

C. $\frac{1}{5}$ and $\frac{1}{4}$

D. $\frac{1}{6}$ and $\frac{1}{5}$

E. $\frac{1}{7}$ and $\frac{1}{6}$

163. For every even positive integer m, $f(m)$ represents the product of all even integers from 2 to m, inclusive. For example, $f(12) = 2 \times 4 \times 6 \times 8 \times 10 \times 12$. What is the greatest prime factor of $f(24)$?

A. 23

B. 19

C. 17

D. 13

E. 11

3, k, 2, 8, m, 3

164. The arithmetic mean of the list of numbers above is 4. If k and m are integers and $k \neq m$, what is the median of the list?

A. 2

B. 2.5

C. 3

D. 3.5

E. 4

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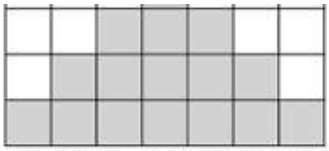

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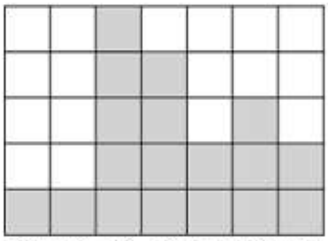

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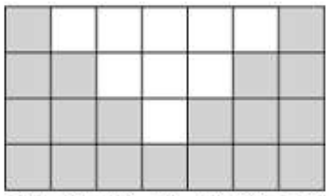

Z

165. If the variables X, Y, and Z take on only the values 10, 20, 30, 40, 50, 60, or 70 with frequencies indicated by the shaded regions above, for which of the frequency distributions is the mean equal to the median?

A. X only

B. Y only

C. Z only

D. X and Y

E. X and Z

$$2x + y = 12$$
$$|y| \leq 12$$

166. For how many ordered pairs (x, y) that are solutions of the system above are x and y both integers?

A. 7

B. 10

C. 12

D. 13

E. 14

167. The United States Mint produces coins in 1-cent, 5-cent, 10-cent, 25-cent, and 50-cent denominations. If a jar contains exactly 100 cents worth of these coins, which of the following could be the total number of coins in the jar?

I. 91

II. 81

III. 76

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

168. A certain university will select 1 of 7 candidates eligible to fill a position in the mathematics department and 2 of 10 candidates eligible to fill 2 identical positions in the computer science department. If none of the candidates is eligible for a position in both departments, how many different sets of 3 candidates are there to fill the 3 positions?

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positions?

A. 42

B. 70

C. 140

D. 165

E. 315

169. A survey of employers found that during 1993 employment costs rose 3.5 percent, where employment costs consist of salary costs and fringe-benefit costs. If salary costs rose 3 percent and fringe-benefit costs rose 5.5 percent during 1993, then fringe-benefit costs represented what percent of employment costs at the beginning of 1993?

A. 16.5%

B. 20%

C. 35%

D. 55%

E. 65%

170. The subsets of the set $\{w, x, y\}$ are $\{w\}$, $\{x\}$, $\{y\}$, $\{w, x\}$, $\{w, y\}$, $\{x, y\}$, $\{w, x, y\}$, and { } (the empty subset). How many subsets of the set $\{w, x, y, z\}$ contain w?

A. Four

B. Five

C. Seven

D. Eight

E. Sixteen

171. The number $\sqrt{63 - 36\sqrt{3}}$ can be expressed as $x + y\sqrt{3}$ for some integers x and y. What is the value of xy?

A. –18

B. –6

C. 6

D. 18

E. 27

172. There are 10 books on a shelf, of which 4 are paperbacks and 6 are hardbacks. How many possible selections of 5 books from the shelf contain at least one paperback and at least one hardback?

A. 75

B. 120

C. 210

D. 246

E. 252

173. If x is to be chosen at random from the set {1, 2, 3, 4} and y is to be chosen at random from the set {5, 6, 7}, what is the probability that xy will be even?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{1}{2}$

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2

D. $\frac{2}{3}$

E. $\frac{5}{6}$

174. The function f is defined for each positive three-digit integer n by $f(n) = 2^x\ 3^y\ 5^z$, where x, y, and z are the hundreds, tens, and units digits of n, respectively. If m and v are three-digit positive integers such that $f(m) = 9f(v)$, then $m - v =$

A. 8

B. 9

C. 18

D. 20

E. 80

175. If $10^{50} - 74$ is written as an integer in base 10 notation, what is the sum of the digits in that integer?

A. 424

B. 433

C. 440

D. 449

E. 467

176. A certain company that sells only cars and trucks reported that revenues from car sales in 1997 were down 11 percent from 1996 and revenues from truck sales in 1997 were up 7 percent from 1996. If total revenues from car sales and truck sales in 1997 were up 1 percent from 1996, what is the ratio of revenue from car sales in 1996 to revenue from truck sales in 1996?

A. 1:2

B. 4:5

C. 1:1

D. 3:2

E. 5:3

177. Becky rented a power tool from a rental shop. The rent for the tool was \$12 for the first hour and \$3 for each additional hour. If Becky paid a total of \$27, excluding sales tax, to rent the tool, for how many hours did she rent it?

A. 5

B. 6

C. 9

D. 10

E. 12

178. If $4 < \frac{7-x}{3}$, which of the following must be true?

I. $5 < x$

II. $|x+3| > 2$

III. $-(x+5)$ is positive.

A. II only

B. III only

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B. III only

C. I and II only

D. II and III only

E. I, II, and III

179. On a certain day, a bakery produced a batch of rolls at a total production cost of $300. On that day, $\frac{4}{5}$ of the rolls in the batch were sold, each at a price that was 50 percent greater than the average (arithmetic mean) production cost per roll. The remaining rolls in the batch were sold the next day, each at a price that was 20 percent less than the price of the day before. What was the bakery's profit on this batch of rolls?

A. $150

B. $144

C. $132

D. $108

E. $90

180. A set of numbers has the property that for any number t in the set, $t + 2$ is in the set. If –1 is in the set, which of the following must also be in the set?

I. –3

II. 1

III. 5

A. I only

B. II only

C. I and II only

D. II and III only

E. I, II, and III

181. A couple decides to have 4 children. If they succeed in having 4 children and each child is equally likely to be a boy or a girl, what is the probability that they will have exactly 2 girls and 2 boys?

A. $\frac{3}{8}$

B. $\frac{1}{4}$

C. $\frac{3}{16}$

D. $\frac{1}{8}$

E. $\frac{1}{16}$

182. The closing price of Stock X changed on each trading day last month. The percent change in the closing price of Stock X from the first trading day last month to each of the other trading days last month was less than 50 percent. If the closing price on the second trading day last month was $10.00, which of the following CANNOT be the closing price on the last trading day last month?

A. $3.00

B. $9.00

C. $19.00

D. $24.00

E. $29.00

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E. $29.00

183. How many different positive integers are factors of 441?

A. 4

B. 6

C. 7

D. 9

E. 11

184. If n is a positive integer and n^2 is divisible by 72, then the largest positive integer that must divide n is

A. 6

B. 12

C. 24

D. 36

E. 48

185. A certain grocery purchased x pounds of produce for p dollars per pound. If y pounds of the produce had to be discarded due to spoilage and the grocery sold the rest for s dollars per pound, which of the following represents the gross profit on the sale of the produce?

A. $(x-y)s-xp$

B. $(x-y)p-ys$

C. $(s-p)y-xp$

D. $xp-ys$

E. $(x-y)(s-p)$

186. If x, y, and z are positive integers such that x is a factor of y, and x is a multiple of z, which of the following is NOT necessarily an integer?

A. $\frac{x+z}{z}$

B. $\frac{y+z}{x}$

C. $\frac{x+y}{z}$

D. $\frac{xy}{z}$

E. $\frac{yz}{x}$

187. Running at their respective constant rates, Machine X takes 2 days longer to produce w widgets than Machine Y. At these rates, if the two machines together produce $\frac{5}{4}w$ widgets in 3 days, how many days would it take Machine X alone to produce $2w$ widgets?

A. 4

B. 6

C. 8

D. 10

E. 12

188. What is the greatest positive integer n such that 5^n divides $10!-(2)(5!)^2$?

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A. 2

B. 3

C. 4

D. 5

E. 6

189. Yesterday, Candice and Sabrina trained for a bicycle race by riding around an oval track. They both began riding at the same time from the track's starting point. However, Candice rode at a faster pace than Sabrina, completing each lap around the track in 42 seconds, while Sabrina completed each lap around the track in 46 seconds. How many laps around the track had Candice completed the next time that Candice and Sabrina were together at the starting point?

A. 21

B. 23

C. 42

D. 46

E. 483

190. If $n = 9! - 6^4$, which of the following is the greatest integer k such that 3^k is a factor of n?

A. 1

B. 3

C. 4

D. 6

E. 8

191. The integer 120 has many factorizations. For example, 120 = (2)(60), 120 = (3) (4)(10), and 120 = (–1)(–3)(4)(10). In how many of the factorizations of 120 are the factors consecutive integers in ascending order?

A. 2

B. 3

C. 4

D. 5

E. 6

192. Jorge's bank statement showed a balance that was \$0.54 greater than what his records showed. He discovered that he had written a check for $\$x.yz$ and had recorded it as $\$x.zy$, where each of x, y, and z represents a digit from 0 though 9. Which of the following could be the value of z?

A. 2

B. 3

C. 4

D. 5

E. 6

193. One side of a parking stall is defined by a straight stripe that consists of n painted sections of equal length with an unpainted section $\frac{1}{2}$ as long between each pair of consecutive painted sections. The total length of the stripe from the beginning of the first painted section to the end of the last painted section is 203 inches. If n is an integer and the length, in inches, of each unpainted section is an integer greater than 2, what is the value of n?

A. 5

B. 9

C. 10

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C. 10

D. 14

E. 29

194. This year, Henry will save a certain amount of his income, and he will spend the rest. Next year, Henry will have no income, but for each dollar that he saves this year, he will have $1 + r$ dollars available to spend. In terms of r, what fraction of his income should Henry save this year so that next year the amount he has available to spend will be equal to half the amount that he spends this year?

A. $\frac{1}{r} + 2$

B. $\frac{1}{2r} + 2$

C. $\frac{1}{3r} + 2$

D. $\frac{1}{r} + 3$

E. $\frac{1}{2r} + 3$

Machine	Consecutive Minutes Machine Is Off	Units of Power When On
A	17	15
B	14	18
C	11	12

195. At a certain factory, each of Machines A, B, and C is periodically on for exactly 1 minute and periodically off for a fixed number of consecutive minutes. The table above shows that Machine A is on and uses 15 units of power every 18th minute, Machine B is on and uses 18 units of power every 15th minute, and Machine C is on and uses 12 units of power every 12th minute. The factory has a backup generator that operates only when the total power usage of the 3 machines exceeds 30 units of power. What is the time interval, in minutes, between consecutive times the backup generator begins to operate?

A. 36

B. 63

C. 90

D. 180

E. 270

196.

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